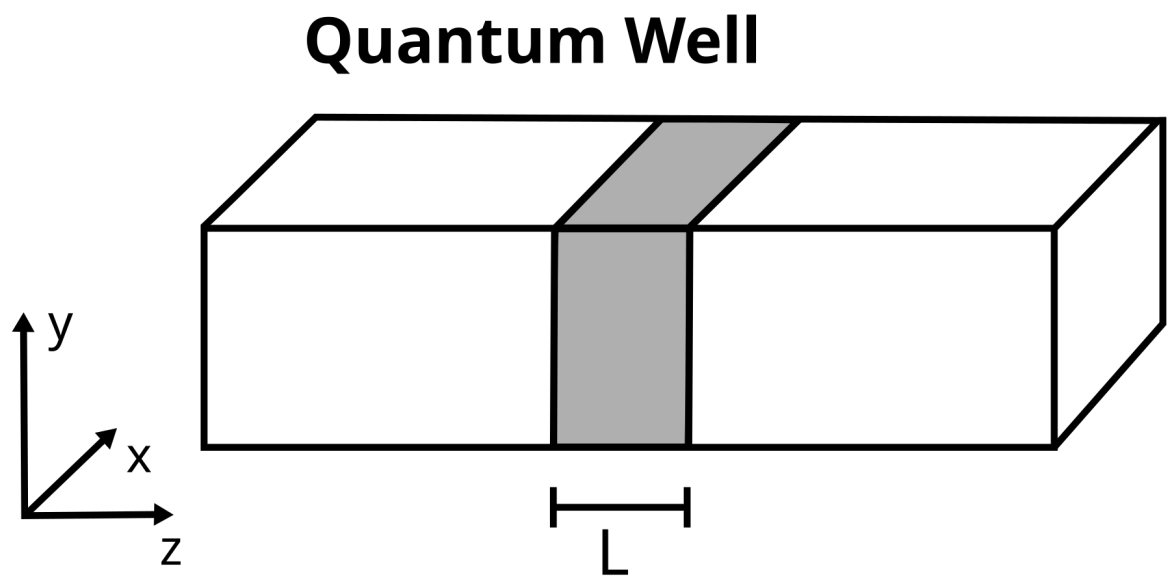
EE2021: Assignment -3 (Hardcopy submission by November 10th, 2025)

1. A 2D quantum well is given where electrons are free to move in X direction. Follow the methodology used for 3D density of states (DoS) in class, derive and show that the DoS expression for this system is

$$g_{2D}(E) = \frac{m^*}{\pi\hbar^2}$$



Consider the well as an infinite potential well, find the lowest two eigenvalues for L=5 nm. The bias conditions are such that the position of chemical potential (Fermi level) is at the midway between these two eigenvalues. Take $m^*=0.5m_0$. Derive the expression for the density of electrons at both energy levels and find their occupation probability.